

Patterns of Exploratory Behavior in the Spiny Mouse, *Acomys cahirinus*

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The present paper reports an investigation of patterns of exploratory behavior shown by laboratory-bred Spiny Mice (*Acomys cahirinus*) when given access to a large, novel arena. The aim was to test hypotheses suggested by our previous work with this species. Previous experiments in which the exploratory behavior of *Acomys* was compared with that of *Mus* had suggested that each species had a characteristic pattern of emergence and exploration. The present experiment addressed the question of whether the exploratory patterns of *Acomys* could be predicted from patterns of emergence as suggested by earlier experiments. Data are presented which indicate that the exploratory behavior of *Acomys* in a large, novel arena is reasonably predictable. The data presented include measures of the animal's responses to different stimuli within the novel environment, including novel, conspicuous objects, food sources, and the holding cage to which the animal could return. In addition, the patterns of movement through different areas of the environment were recorded in relation to the behavior of the animal when it was first given access to the arena. The data indicate that there are two types of strategy, or patterns, by which *Acomys* begin to explore a novel environment. The first type consists of a delayed emergence into the arena, followed by brief excursions into the area immediately adjacent to the holding cage, interspersed with longer periods of returns to the holding cage. The second type involves immediate emergence, followed by rapid "dashes" around the periphery of the arena. Which of the two strategies is adopted appears to depend upon the animal's behavior at the time of initial access. Two subsequent experiments considered these strategies further. Experiment 2 indicated that Type I was more characteristic of males and Type II more characteristic of females, although there were no differences related to the female estrous cycle. In Experiment 3, the patterns of exploration over four consecutive tests were investigated. It was found that the strategy adopted by an individual is likely to be consistent across tests. © 1986 Academic Press, Inc.

When faced with an unfamiliar environment or with changes in a familiar one, animals typically show behavior that we might term exploratory, such as orienting to, sniffing, or manipulating various stimuli.

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In all these events we might infer that they were investigating and gaining information about their surroundings. They might also show other responses, less readily labeled as exploratory, such as scent-marking, running away, or immobility.

In previous studies of exploratory behavior we have begun to compare some patterns of exploration in unfamiliar environments shown by the Spiny Mouse, *Acomys cahirinus*, with those of the more extensively studied laboratory mouse, *Mus musculus* (Birke, D'Udine, & Albonetti, 1985). The Spiny Mouse is a small, murid rodent originating in semiarid zones of the Near and Middle East (Dieterlen, 1962). They are increasingly being used in laboratory studies of behavior, particularly because they are unique among the Muridae in giving birth to precocial young. Little is known, however, of their patterns of exploratory behavior, either in the wild or in captivity, although they show great readiness to explore novel objects under laboratory conditions (personal observation).

Acomys was used in the present series of experiments in the context of a longer-term research program investigating the development of behavioral strategies of exploration in which the precocial young of *Acomys* can be used from the day of birth. A principal aim of the present study was thus to outline in somewhat descriptive terms the overall patterns by which adults of this species set about exploring unfamiliar areas, in order to provide a descriptive base for further investigation of behavioral development. It was also designed to test some specific predictions about patterns of exploration arising from our previous experiments with adult *Acomys*.

The general strategy of exploration employed by any given species is likely to reflect a balance between the costs, such as predation risk, and the benefits in terms of information gained. In one of the few studies to look at predation risk in relation to exploratory behavior, Glickman and Morrison (1969) found that mice which were particularly exploratory in an open field were those most likely to be preyed upon, whereas those that remained close to the edge and moved little were more likely to survive. This kind of predator selection is clearly likely to favor strategies of wall-seeking behavior by small rodents rather than one of forays into open areas. Moreover, an animal that encounters a new environment knows little about it, including the location of important resources. An optimal strategy of exploration might thus be for the animal to explore the area in a general way for a while, possibly sampling different patches of resources, before searching more specifically within a patch when the general topography was known. This appears to be the way in which some animals do explore unfamiliar areas. Balda (1980) observed, for example, that nutcrackers (*Nucifraga caryocatactes*) first carry out a general survey of an area, before searching specifically for a cache site.

The strategies of exploration employed by animals in unfamiliar en-

vironments is perhaps most easily investigated under captive, controlled conditions, and the present experiments were designed to test very general predictions. From the foregoing, it might be predicted that a small rodent would tend initially to use wall-seeking behavior as a means of carrying out a general survey, before investigating specific stimuli in the novel environment more closely. Our previous experiments had indicated that *Acomys* typically emerge rapidly into an arena and then show a high level of locomotor exploration around the periphery of the arena. The first hypothesis therefore concerned changes in exploratory behavior over time; that is, as the novel environment becomes familiar, over a period of time, locomotor exploration would decrease, while exploration of specific stimuli would increase.

The second hypothesis was concerned with individual differences in exploratory behavior. Although the strategy employed by *Acomys* appeared from our previous study to be one of rapid emergence followed by fast movement around the perimeter, this did not seem to be the pattern employed by *Mus*. Most of the *Mus* tested, as well as a few *Acomys*, seemed to be much more cautious about emergence. They were considerably slower to leave the start box, and then tended to make short initial forays into the arena, rather than running around it, combined with frequent returns to the start box. This more cautious approach appeared to continue throughout the testing period.

Both types of response might equally well provide the animal with initial information, although the strategy employed to gain it seems to differ. Approximately one quarter of all *Acomys* tested showed this more cautious behavior with long emergence latencies (over 60 s) and frequent returns. The remainder all showed the fast emergence and rapid dashes described above. The experiments reported here also addressed the question of whether these differences in apparent strategy reflected transient differences in motivational state, or differences between individuals and their preferred exploratory pattern; they were thus also designed to test the hypothesis that these differences were features of particular individuals. If so, they would be expected to be relatively consistent over time, either within a testing period, or between consecutive tests. This prediction was tested in the experiments reported here.

EXPERIMENT 1

Methods

Forty adult Spiny Mice were used in this study, 17 males and 23 females. They were bred from stock maintained at the Open University, and derived from original stock kindly donated by the Zoological Society of London. Animals were put into single sex groups of 4–6 as soon as they were weaned, and housed in standard laboratory rat cages. Water and food (rat chow and sunflower seeds) were available ad libitum, and

a supplement of small pieces of apple was provided twice weekly. All animals were maintained and tested under a reversed lighting schedule, with lights on from 2100 to 0900 h. Dim red lights provided background illumination during the dark period and during testing.

Prior to testing, the animal was removed from the group cage and placed singly in an holding cage. As the holding cage was identical in size and shape to the home cage, it was considered to provide a standard stimulus, while being relatively familiar to the animal, by comparison with the novel arena.

The animal's holding cage was joined to a door into the arena by a connecting tube 20 cm in length made of aluminum. On one end, it connected via a sliding door into the arena, and at the other, it connected to the cage via an aperture adjacent to the food hopper. After a 5-min period to allow the animal to familiarize itself with the holding cage, the sliding door into the arena was removed, and recording began. All movements of the animal in the arena over the subsequent 2 h were recorded on videotape using a Panasonic video cassette recorder, coupled with a camera placed above the arena and capable of operating at low light intensities. As soon as the period of observation was completed the animal was removed, returned to its home cage, and the entire arena washed down with a dilute detergent solution.

The arena itself was rectangular and made of aluminum, $120 \times 120 \times 30$ cm. The floor was made of clear Perspex. At a distance of 30 cm from each corner, along the diagonal, was a small object comprising a small disc made of black Perspex. This was held in place by a bolt passing through the floor. At a distance of 40 cm from each corner were small glass beakers, set into holes in the floor, so that the rims were flush with the floor surface. One of these was empty, and the others contained food sources. The first contained sunflower seeds, and the second contained rat pellets, which are both food types normally available to the Spiny Mice in our colony. The fourth beaker contained a powdered food (CDXM primate diet). All these foods will be eaten by Spiny Mice in their home cages, although sunflower seeds are always preferred (personal observations). Each beaker was half-filled with the food source and placed into one of the holes. The specific location of each food source was varied randomly between animals to avoid location effects.

In order to analyze the large amount of material on the videotapes, it was necessary to sample the behavior in some way. This was done as follows:

(i) First, in order to analyze changes over time, 2-min intervals were sampled at the beginning of each 20-min period for each animal. During these six sampling periods, the following measures were taken from the video records, and transferred to computer disc, using an Apple II microcomputer, for later analysis.

(a) *Activity*. A score was taken of the number of imaginary grid squares crossed during each sampling period (the grid being based on a division into 25 squares).

(b) *Cage use*. The duration of use of the holding cage was recorded. Since this was off the video screen, this provides no record, however, of the animal's behavior while in the holding cage.

(c) *Rearing*. The frequency of rearing behavior was recorded.

(d) *Food sources*. The frequency of visits to each food source and the duration of time during which the animal's head was inside any of the beakers were scored.

(e) *Objects*. The frequency of visits to the objects was scored.

(ii) Second, the patterns of movement during the initial 20-min period were measured to test the predictions noted about behavior following initial emergence. This was done by measuring the frequency of square entry using the 25-square grid noted above. These were labeled using a grid drawn onto the television monitor. Square entries for each 5-min period of the initial 20 minutes were recorded.

Not all the data were used for this analysis, however. Preliminary inspection of initial emergence times for all 40 animals indicated a bimodal distribution, with 25 animals having emergence times of 10 s or less and 10 having emergence times between 90 and 120 s. The remaining 5 were distributed between these extremes. On the basis of these scores, an arbitrary selection was made, such that the 10 animals with long emergence times were matched at random to 10 animals showing very short emergence times. This arbitrary selection on the basis of emergence times allowed a direct contrast to be made to test the second hypothesis above—that is, that animals showing delayed emergence would show different patterns of exploratory behavior to those emerging quickly.

Results

(i) *Analysis from whole trial data*. The data from various measures were analyzed using two- or three-way analyses of variance with time periods and sex as variates. Some of the data obtained from these measures are shown in Fig. 1 and Table 1 which give the mean and standard error for duration of holding cage use $\times$ time period, activity scores, rearing frequencies, the frequency of visits to objects, and the duration of time spent at the different food sources.

Rearing frequency showed a significant effect due to sex of animal, with females consistently showing higher levels of rearing than males at all time periods ($F(1, 33) = 3.77; p < .05$). There was no significant effect of time period ($F = 0.49$). The frequency of visits to the objects was significant for time period ($F = 3.89; p < .05$), but not for sex ($F = 2.83$; ns). For both frequency and duration measures of food sources, there were effects of food type (frequency: $F = 11.2, p < .01$; duration:

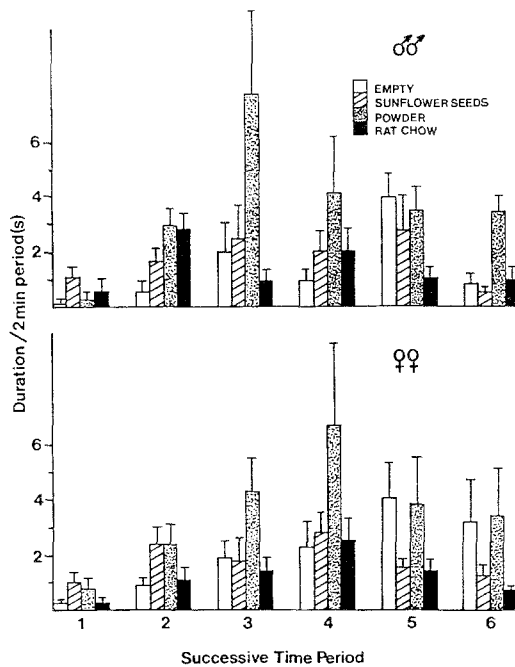


FIG. 1. Mean duration of time spent visiting each food source in successive time periods. Open bars show the mean time visiting the empty beaker ($\pm$ SEM); hatched bars show the time visiting beakers containing sunflower seeds; the dotted bars show the time spent visiting the beaker containing powdered primate diet; and the black bars show the mean time spent visiting the beaker containing pelleted rat chow.

$F = 13.2$, $p < .01$) and of time period (frequency: $F = 12.2$, $p < .01$; duration: $F = 8.5$, $p < .01$). There was also a significant interaction between food type and time period (frequency: $F = 2.97$, $p < .05$; duration: $F = 2.93$, $p < .05$). There were no other significant effects from these analyses.

(ii) *Patterns of movement in initial period of exposure.* Before analysis, the data were divided, as noted above, according to the initial response of the animal to the arena, into Type I (delayed emergence) and Type II (immediate emergence). The analysis of patterns of movement during the first 20 min of testing gave results that are summarized in Tables 2 and 3. Table 2 gives the mean ($\pm$ SEM), frequency, and duration of entry into particular parts of the arena, and Table 3 gives some of the results of the analysis in terms of a transition matrix. For simplicity, not all grid squares were treated separately, however. Rather, the tables give results for holding cage entry, entry to the squares adjacent to the holding cage, and entry to the center of the arena. In addition, it gives scores for entries into the two corners nearest the holding cage and into the two corners farthest from the holding cage. Similarly, Table 3 gives only

TABLE 1
Behavior in Arena over Whole Testing Period (Mean $\pm$ SEM)

Behavior	Time Period					
	1	2	3	4	5	6
Duration in holding cage (S)						
Males	41.6 $\pm$ 7.3	36.4 $\pm$ 7.0	52.3 $\pm$ 14.0	45.4 $\pm$ 10.4	55.7 $\pm$ 12.3	66.7 $\pm$ 19.4
Females	66.4 $\pm$ 13.6	47.7 $\pm$ 10.1	46.6 $\pm$ 7.2	54.6 $\pm$ 9.9	53.4 $\pm$ 9.6	68.8 $\pm$ 12.0
Activity score						
Males	26.2 $\pm$ 2.9	31.9 $\pm$ 2.8	27.8 $\pm$ 3.8	31.9 $\pm$ 3.8	29.9 $\pm$ 3.1	29.3 $\pm$ 5.1
Females	22.7 $\pm$ 3.2	31.3 $\pm$ 2.9	32.9 $\pm$ 2.4	31.0 $\pm$ 2.4	32.4 $\pm$ 3.3	24.1 $\pm$ 3.3
Rearing						
Males	2.0 $\pm$ 1.3	5.4 $\pm$ 1.9	4.0 $\pm$ 1.8	3.6 $\pm$ 1.2	3.4 $\pm$ 1.6	4.4 $\pm$ 1.7
Females	4.5 $\pm$ 1.3	7.7 $\pm$ 1.8	7.9 $\pm$ 1.8	6.7 $\pm$ 2.0	9.0 $\pm$ 2.5	5.5 $\pm$ 1.5
Visits to objects						
Males	3.8 $\pm$ 0.3	3.95 $\pm$ 0.3	4.4 $\pm$ 0.3	4.8 $\pm$ 0.3	4.45 $\pm$ 0.23	3.25 $\pm$ 0.24
Females	2.0 $\pm$ 0.1	3.9 $\pm$ 0.15	3.75 $\pm$ 0.2	3.75 $\pm$ 0.25	2.9 $\pm$ 0.2	3.0 $\pm$ 0.27

TABLE 2
Patterns of Movement

Location	Type I Time Period					Type II Time Period			
	1	2	3	4		1	2	3	4
Frequency, holding cage	8.9 ± 1.2	7.0 ± 0.8	6.4 ± 0.7	5.3 ± 0.7		7.8 ± 0.8	5.6 ± 0.5	5.3 ± 0.5	5.6 ± 0.6
Adjacent square	16.1 ± 2.1*	17.5 ± 3.1	16.5 ± 2.5	14.7 ± 2.2		21.4 ± 1.1*	19.0 ± 1.5	14.6 ± 1.1	16.9 ± 1.1
Central square	0.14 ± 0.1**	1.5 ± 0.8	1.9 ± 0.6*	2.5 ± 0.6		2.8 ± 0.8**	3.9 ± 0.7*	3.7 ± 0.4*	3.7 ± 0.8
Nearest corners	3.7 ± 1.6**	10.6 ± 3.7	14.7 ± 3.9	13.5 ± 2.9		15.8 ± 2.0**	16.7 ± 1.6	9.7 ± 0.9	13.8 ± 0.6
Furthest corners	0.8 ± 0.5**	11.0 ± 3.4	14.9 ± 3.2	11.6 ± 2.9		18.7 ± 1.4**	17.4 ± 1.7	11.1 ± 1.3	10.7 ± 2.2
Duration(s), holding cage	234.6 ± 5.7**	145.3 ± 12.8	115.7 ± 20.4	125.0 ± 20.0		89.1 ± 7.8**	97.1 ± 29.1	134.5 ± 19.8	133.7 ± 24.1
Adjacent square	27.7 ± 4.3	26.3 ± 4.9	25.1 ± 3.4	33.1 ± 6.7		30.6 ± 2.9	28.7 ± 4.7	23.4 ± 3.0	29.4 ± 4.6
Central square	0.3 ± 0.3*	1.6 ± 1.0	2.0 ± 0.7	3.0 ± 1.0		1.7 ± 0.5*	2.8 ± 0.7	1.8 ± 0.3	2.2 ± 0.4
Nearest corners	11.7 ± 4.6**	21.4 ± 5.8	33.3 ± 5.0	29.1 ± 6.9		34.4 ± 5.2**	28.1 ± 3.9	22.2 ± 3.2	25.7 ± 2.8
Furthest corners	2.8 ± 1.8**	26.2 ± 9.0	38.3 ± 7.1	38.8 ± 10.8		52.6 ± 6.9**	36.8 ± 6.1	24.0 ± 2.6	21.4 ± 4.1

Note. Values = Mean ± SEM.

* Values significantly different for that time period at $p \leq .05$.** Values significantly different for that time period at $p \leq .01$.

TABLE 3
Patterns of Movement: Transition Frequencies

Movement through	% transitions (first 5 min)	
	Type I	Type II
Nearest side	44.9	20.3
Farthest side	1.7	19.7
Parallel sides	6.3	14.3
	7.0	18.2
Home cage $\rightleftharpoons$ nearest squares	38.6	7.4
Total no. transitions	285	1386

transition frequencies for the four sides of the arena, and into or out of the holding cage. Because measures of duration were not normally distributed, these scores were log-transformed prior to analysis.

Data were compared for the two groups across consecutive time periods using a two-way analysis of variance with repeated measures. In order to analyze separately the behavior of the animals when first given access, data for the first 5-min period were also analyzed alone using the *t* test.

On duration measures, the ANOVAs showed a significant difference between the two groups for holding-cage use ($F_{1,18} = 4.8$, $p < .05$) and a significant interaction on this measure with time period ($F_{3,18} = 10.9$, $p < .01$). For durations in the nearest and farthest corners, respectively, there was no significant effect of either group or time period ($F < 1.7$), but there were significant interactions ($F_{3,54} = 5.0$ and 11.4 , respectively; $p < .01$). No significant differences were found on duration measures in either the central or the adjacent square.

On frequency measures, there was a significant effect of time period on holding-cage use ($F_{3,18} = 7.2$, $p < .01$) and on adjacent square use ($F_{3,18} = 3.4$, $p < .05$). There was a significant effect of both group ($F_{1,18} = 9.3$, $p < .01$) and time period ($F_{3,18} = 3.15$; $p < .05$) on frequency of entry to the central square. For frequency of entry to the nearest and farthest corners, only the interaction terms were significant ($F_{3,54} = 11.9$ and 8.3 , respectively; $p < .01$).

Inspection of the data in Table 2 indicates that the greatest difference between the two groups occurs in the first 5 min. Analysis of these data alone showed significant differences on several measures (*t* test): duration of holding-cage use ($p < .05$); duration in central square, nearest and farthest corners (all $p < .01$); frequency of entry to the adjacent square to the holding cage ($p < .05$); and to the corners ($p < .01$) and to the central square ($p < .01$).

The data shown in Table 3 also indicate the difference in patterns of

movement between the two types. Type I animals, for example, move through the squares nearest to the holding cage a great deal, with 44.9% of all their transitions into and out of the holding cage; thus, 38.6% of their transitions are between the holding cage and the adjacent three squares. By contrast, Type II animals show many more transitions along the far edge, i.e., 19.7% transitions along that edge, compared to 1.7% for Type I animals.

DISCUSSION

These results indicate some of the ways in which Spiny Mice begin to explore an unfamiliar environment. The first part of the analysis concerned the way in which various measures of behavior changed at different sampling times during the whole 2-h period. The hypothesis was that the strategy adopted by the animals would be an initial period of general survey, followed by more detailed examination of specific stimuli. From this, a gradual decline in locomotor activity over time might be predicted, accompanied by a gradual increase in inspection of stimuli within the central area. However, neither holding-cage use nor activity scores showed significant changes over time, suggesting that the animals showed no tendency toward a decrease in locomotor exploration over the 2-h test period. Rearing behavior also did not vary consistently with time, although it did show a sex difference. This is consistent with findings from other rodents, such as the rat. A number of studies have shown that female rats are more active in the open field, including showing more rearing behavior, than are males (e.g., Stevens & Goldstein, 1981). Sex differences in rearing behavior in *Acomys* have not been systematically investigated, although Porter (1976) noted that female *Acomys* were more active than males when tested in social situations. Rearing has been considered to be a form of exploratory behavior, and might be expected to show a positive correlation with activity levels (Lat, 1963), so it is surprising that, in this experiment, sex differences were found for rearing but not for activity.

Although the results from activity measures do not support the predictions made about change over time, the behavior of the animals toward other stimuli did change as predicted. Both visits to the plastic objects and visits to food sources did show an increase over the first three or four time periods, although they declined subsequently. These data are discussed in more detail below (see General Discussion).

The second set of data presented here concerns the behavior of animals following initial emergence. We noted above that animals differed in the responses they made when first given access. The question being addressed by this analysis was whether or not these animals differed not only in their initial emergence, but also in the way that they explored subsequently. It is clear from the results that, at least in the first 5 min, the two groups

did behave quite differently in the arena, although the difference declined in subsequent time periods. Thus, the behavior of the animal when it is first given access to an unfamiliar area seems to yield some predictability about its ensuing behavior. Specifically, Type II animals adopted the strategy outlined in the introductory remarks; that is, they made rapid dashes around the periphery of the arena. Type I animals seemed on the other hand to adopt a "cautious" strategy, making short forays into the arena followed by returns to the relative safety of the holding cage.

Although specific behavior patterns were not scored in association with the collection of these movement data, Type I animals appeared to adopt the posture that Van der Poel (1979) described for the rat as a "stretched attention" posture, in which the hindlimbs and snout are moved tentatively forward, associated with sniffing and small vibrissal movements, while the hindlimbs remain stationary. This posture appeared to be associated only with the short forays shown initially by Type I animals, and may be described as indicating ambivalence (Van der Poel, 1979).

The second approach, characterised by Type II animals, is to emerge rapidly, without any characteristic posture, and to explore the arena initially by running fast around the boundary walls. Type II animals not only spend more time in the arena during the first time period, but are also more active while they are there, making an average of 66 transitions per minute spent in the arena, compared to 41 for Type I animals.

The results of this experiment suggest that there are two strategies or patterns by which *Acomys* explore novel environments. These differ mainly in the initial phase of exposure, while the animal is presumably gaining general information about the topography of the area. In the next two experiments, the possibility was considered that these differences reflect differences among individuals rather than differences due to transient motivational states at the time of testing.

The Type I group in the previous experiment consisted of six males and four females, but it was not clear whether the two strategies were sex-related characteristics. In the next experiment, the exploratory behavior of a larger group of animals in the same apparatus was investigated in order to determine whether the sex bias in group composition noted in Experiment 1 was representative. That is, are different strategies of exploration employed by animals of each sex? Given that changes in exploratory behavior have been shown to occur with the estrous cycle in other rodents (e.g., rats, Birke & Archer, 1975, and Birke, 1979; mice, Wolff & Powell, 1981), it is possible that the estrous stage might affect any transient motivational state. Thus, females were tested at both estrus and diestrus in this experiment, which used a simplified design based on the apparatus used in the first experiment.

EXPERIMENT 2

Sex Differences in Acomys Exploratory Behavior Methods

Twenty male and 30 female *Acomys* were used in this experiment, housed and maintained as described for experiment one. The females' estrous cycles were monitored using vaginal smears (see Peitz, 1981), so that 15 were tested when in estrus and 15 when in diestrus.

For testing, animals were placed in a holding cage and subsequently allowed access to the arena, as described above. There were two important differences from the first experiment, however. In the first place, animals were only tested for 10 min after emergence since the results of Experiment 1 indicated that individual differences declined after this period. Secondly, the beakers were left empty, so that any responses to them could be classed as exploratory.

Behavior was scored as described above using the following categories:

(a) *Frequencies*. The frequencies of rearing, object investigation, entry to the holding cage, and squares crossed were scored for each animal. In addition, the frequency of head-dipping into the empty beakers was scored.

(b) *Duration*. The duration of time spent in the holding cage was scored, as well as the latency for the animal to emerge from the holding cage when first allowed access.

Results. Four males did not emerge at all during the 10-min period after being given access, so they were dropped from further analysis.

The results of this experiment are summarized in Table 4. Data were analyzed for each measure using one-way ANOVAs. This showed a significant sex difference for rearing ($F_{2,44} = 9.565$, $p < .01$), for object investigation ($F = 4.1$, $p < .05$), for head-dipping ($F = 8.44$, $p < .01$). The number of squares crossed did not show a sex difference ($F = 3.0$, ns), although the difference is only just nonsignificant.

There was also a significant sex difference in the duration spent in the

TABLE 4
Comparison of the Exploratory Behavior of Male and Female *Acomys*

Measure	Sex		
	Male	Estrous female	Diestrous female
Rearing	12.9 $\pm$ 2.9	30.5 $\pm$ 2.6	39.5 $\pm$ 7.5
Object investigation	14.5 $\pm$ 3.3	21.5 $\pm$ 2.6	26.4 $\pm$ 3.1
Head-dips	2.9 $\pm$ 0.8	6.5 $\pm$ 0.9	8.3 $\pm$ 1.2
Squares cross (activity)	237.8 $\pm$ 31.7	320.0 $\pm$ 41.6	349.9 $\pm$ 61.8
Frequency holding-cage entry	4.9 $\pm$ 0.8	9.9 $\pm$ 0.9	10.5 $\pm$ 1.2
Duration holding cage (s)	438.0 $\pm$ 33.4	281.3 $\pm$ 21.6	243.9 $\pm$ 37.8
Emergency latency (s)	279.5 $\pm$ 57.4	59.4 $\pm$ 31.0	62.8 $\pm$ 31.0

holding cage ($F_{1,48} = 9.3, p < .01$; log-transformed data). For all of these measures, however, post hoc Duncan tests showed that the difference was between males and females rather than between females in different estrous states ($p < .05$). This is evident from the data presented in Table 4.

Discussion

The results of this experiment suggest that the distinction made in Experiment 1 between the two types of exploratory strategy is associated with differences between males and females. In general, males are slower to emerge and spend more time in the holding cage than do estrous or diestrous females. They also show less rearing, a result which agrees with that of Experiment 2, and also less head-dipping than females. In part, these results might be explained by the difference in activity levels (squares crossed) which, although just nonsignificant, was noticeably higher for both groups of females.

However, the differences in emergence latency and holding-cage use suggest that males are indeed more "cautious" than females, conforming more to the Type I outlined above. Females, on the other hand, are more like Type IIs. This was supported: 3 out of 20 males emerged in 20 s or less compared with 7 out of 15 females in each group. Similarly, only 2 females in each group took longer than 2 min to emerge, while 6 males took longer than 2 min, and 4 did not emerge in 10 min.

In general, these results are in agreement with findings of sex differences in locomotor exploration in other species of rodent, such as the rat (see Archer, 1975) in which females tend to explore unfamiliar environments more than males. We did not, however, find any evidence of differences among females with the estrous cycle. This finding does not agree with studies of the rat, for which both heightened locomotor exploration and enhanced object investigation have been reported at estrus (e.g., Martin and Battig, 1980; Birke, 1979).

The results of this experiment, then, suggest that the two types of exploratory pattern may represent a sex difference in response to an unfamiliar environment. Moreover, the lack of variation with the estrous cycle suggests that the differences might not be due to transient motivational changes. This was directly tested in the third experiment which addressed the question, are the two types of response shown over consecutive testing situations, or do they represent transient differences occurring at the time of testing?

EXPERIMENT 3

Exploratory Behavior in Different Tests *Methods*

Thirty-two animals were used for this experiment, 17 males and 15 females. These were selected at random from the stock population according

to the criterion of the two "types" used in Experiment 1. This selection yielded 5 females and 11 males in Type I, and 10 females and 6 males in Type II, from an initial group of 20 animals of each sex.

Animals were given four different tests, separated by 4 days. Each test lasted 10 min after emergence, as before. The tests were as follow:

Test A. The arena was set up as in Experiment 1.

Test B. Same as Test A, but black and white plastic discs, of the same size as those used previously, were substituted for the beakers. There were thus eight objects in this test, affording some visual contrast.

Test C. As Test A, but with the addition of a clear Perspex box in the center square, which just touched the four beakers at its corners. This was 30 cm in height, and had a small hole midway along each face, 2.5 cm in diameter, through which the animal could pass to enter the central square.

Test D. As Test C, but with the addition of a tall cylinder in the center of the Perspex box. This was 10 cm in diameter and 30 cm high and was made of metal painted matte black.

In each test, the following measures were recorded: duration and frequency of entries to the holding cage; frequency of entries to the square adjacent to the cage, to the central square, and to the four corners; and frequency of visits to objects and/or beakers, depending on the test. In Test D, visits to the central object were also recorded. In addition the frequency of squares crossed was scored in each test to provide an approximate activity score.

Results

The results were analyzed using two-way ANOVAs with repeated measures. All measures except the frequency of the holding-cage entry showed significant effects of group and of repeated measures. Only those measures most directly relevant to the hypothesis are presented, however, for the sake of simplicity. Thus, data are shown in Table 5 to summarize results for the frequency of corner entries across tests, overall activity levels, investigation of beakers/inner objects (as a measure of tendency to venture into the central area), and the duration of time spent in the holding cage. Corner use showed significant differences due to group ($F_{3,28} = 6.4, p < .01$) and to the repeated measure ($F_{3,84} = 10.1; p < .01$). Visits to beakers were significantly different from treatment ($F = 5.8; p < .01$) and on the repeated measure ($F = 21.4; p < .01$). Activity levels were significantly different for group ($F = 6.2; p < .01$) and on the repeated measure ($F = 10.8; p < .01$). The amount of time spent in the holding cage was also significantly different for group ($F = 3.24; p < .05$) and for repeated measures ($F = 4.9; p < .05$). The frequency of visits to the central object in test D were analyzed separately, using

TABLE 5
Brief Summary of Exploratory Behavior in Consecutive Tests

Measure	Group	A	B	C	D
Corner entry	I	34.5	47.9	35.0	28.0**
	II	82.0	86.7	51.9	42.2
Beakers	I	2.4	23.0	32.1	18.9**
	II	5.7	24.4	48.1	24.4
Activity in arena	I	82.8	170.2	152.2	181.2**
	II	174.6	221.2	196.5	214.3
Duration in holding cage(s)	I	348.1	237.4	164.5	188.9*
	II	173.8	179.4	141.8	145.8

Note. ** $p < .01$ for both treatment and test (treatment df, 3,28; repeated measure df, 3,84).

* $p < 0.06$. (Data for both sexes combined.)

a one-way ANOVA; this showed no significant group differences ($F_{3,28} = 2.34$; ns).

Discussion

Overall, the results suggest that the distinction between the two types is maintained over different testing situations; significant differences between the groups were observed on most measures. That is, animals that show the more cautious "Type I" strategy on one occasion seem to be likely to do so on another occasion, thus supporting the hypothesis that the adoption of one or the other strategy of exploration is a trait of individuals rather than a consequence of transient motivational states.

The distinction is perhaps clearest on measures of entry to the corners and the duration spent in the holding cage. Type II animals gave the impression that they visited objects incidentally while engaged in rapid locomotor exploration; indeed, they sometimes seemed literally to bump into objects in their path. By contrast, Type I animals appeared to investigate one object and then to switch attention to another, tending to move directly between objects. This general impression is supported by post hoc analyses of transition frequencies between different parts of the arena: Type I animals showed 27.1% of all transitions between objects, compared to 14.5% for Type II animals. Although these experiments were not designed to test such behavior specifically, the data do suggest that what underlies the different strategies may be a difference in the attention given to particular stimuli while exploring.

GENERAL DISCUSSION

These experiments addressed two questions. The first concerned the extent to which exploratory behavior of *Acomys* declined or changed

over time; the second concerned differences in exploratory strategies and the extent to which these were traits of particular individuals. Some measures of exploration did not appear to show any significant decline over the 2-h testing period employed in Experiment 1. These included activity scores, holding cage use, and rearing behavior. Visits to objects and to food sources did, however, change over time.

Perhaps the most interesting results from this part of the analysis are those for visits to food sources. On both frequency and duration measures, there were consistent effects of food type and of time period. During the initial 2-min period, there was very little interest in the food sources, as might be expected. It would be rather risky for an animal to eat from a food source before it had fully explored the environment and determined whether it was hazardous—or of course whether there is a better food source available elsewhere. Use of the food sources rose to a maximum in the third or fourth time sample, and then declined again. Also, there were clear preferences. The beaker containing rat chow was not preferred, and was visited about as often as an empty beaker. Sunflower seeds were preferred slightly more than rat chow, but by far the greatest interest shown was for the powdered diet. This preference was somewhat surprising since, as noted above, *Acomys* were not found to prefer this kind of food in preliminary trials in their home cages. Rather, in the home cage, sunflower seeds were always preferred.

One possible explanation of this change in preference is that the powdered diet is easy to consume. That is, if the animal is to risk eating from food sources that are in relatively open areas, then it might be safer to eat from sources requiring less handling time. On the other hand, the animal might be spending less time at the beakers containing chow or seeds simply because it is taking them out to be consumed elsewhere. Unfortunately, it was not always possible to determine from the videotape records whether this was the case.

It is also noteworthy that the duration of visits to the *empty* beaker rose slightly toward the end of the observation period. This may reflect a tendency for animals to continue to “check” the empty beaker. Cowan (1977) has described the orderly behavior of rats in patrolling familiar areas. When living in a cross-maze, for example, in which one of the arms was empty, rats regularly patrolled even the empty arm. Such regular checking of different areas increases the chances of detecting any changes in the environment, and of finding fresh sources, such as food sites. Bider, Thibault, and Sarrazin (1968) studied the patrolling behavior of raccoons (*Procyon lotor*) and found that, in the early part of the night, raccoons spend time near known food sites. When they have eaten, they move out to spend time in areas offering less reliable supplies of food. If, however, they encounter suitable food in one of these more marginal areas, then they visit that area sooner the following

night. Clearly, the probability of finding new food sources is increased by making regular visits to other sites.

The second part of the study investigated individual differences in exploratory patterns which were distinguished in the first experiment, while subsequent experiments showed that these types were associated with a sex difference, and appeared to be broadly consistent through consecutive tests.

As noted above, locomotor measures have often been used as an index of exploratory activity (but see Berlyne, 1960; Birke and Archer, 1982). On that basis, Type II animals might be said to be more exploratory. Certainly, in the initial period of access, they spend considerably longer than Type I animals moving about the arena and interacting with stimuli in it. A central problem with locomotor measures of exploration is, however, that many other factors may influence an animal's general activity. We cannot necessarily infer that an animal that moves around a great deal is gaining more information about its environment than one that remains in one spot. The latter may well gain information simply by standing still and looking or sniffing. Indeed, Type I animals may well gain as much preliminary information about the arena by making forays into it and then looking out at it from the safety of the holding cage, as the animal that dashes rapidly around the periphery.

These results suggest that the adoption of one or other strategy appears to be a characteristic of that particular individual, being shown in several consecutive tests. The more cautious Type I strategy involves brief initial forays followed by frequent returns to base, and is more typical of males. By contrast, Type II animals begin their exploration by running around the periphery several times at rapid speed. This sort of behavior, more characteristic of females, allows very rapid scanning of stimuli in the environment, but without any sustained attention being given to specific stimuli.

What the functional significance of this sex difference is—if any—remains obscure. It would not appear to be related to sexual motivation per se since there is no obvious change in female behavior with the estrous cycle. It might be related to factors like home range size in the wild, although rather little is known about home range use of wild *Acomys*. It would certainly be interesting to know how male and female *Acomys* use available space in their native habitat.

That these different types of response to novel environments are characteristic of particular individuals also raises the question of potential neural bases. *Acomys* is becoming of increasing interest to neuroanatomists, who have reported an unusually large and well-laminated hippocampus in this species compared to other rodents (D'Udine and Gozzo, 1983). Moreover, the hippocampus is apparently fully laminated at birth (Brunjes, 1984). Given that the hippocampus has been implicated in spatially guided

behavior (Morris, 1982; Olton and Papas, 1980) it is at least possible that observed differences between individuals in patterns of exploratory behavior have some correlate in, for example, hippocampal function. Whether this is so, and how it develops in this precocial murid, are questions that await further research.

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